

# Shadow run against the verified assessment — Egypt

EGY\_shadow\_g2 compared row by row with EGY\_v17, the verified assessment. The verified assessment was not read by the pipeline and has not been modified by this comparison.

## The five questions

- 1. How many of the 57 rows land on the same level? 29 of 57 (51%).** Of the 32 rows where both assessments set a level at all, 20 agree exactly (62%) and 27 are within one level (84%).
- 2. Do all twelve prerequisites match? 8 of 12** carry the same status. The divergences are listed below; each one moves at least one column of the readiness matrix.
- 3. Gaps.** The verified assessment records 5 (2.1, 5.12, 6.3, 6.13, 8.17). The shadow run records 14, of which it found 5 of the recorded ones and raised 9 the verified assessment does not carry. It also set 10 ratification holds against the verified assessment's 5.
- 4. The rural/national trap at 2.1. Avoided.** The shadow run recorded 2.1 as Gap with no level; gate verdict gap. It did not record a national coverage figure against an indicator naming rural coverage — the error that passed an assessor gate and an initial peer review before an audit caught it.

This country's verified assessment records 2.1 as Gap at no level, so the two agree.

On the construct: *The indicator names RURAL mobile broadband coverage. The best available Egypt-specific evidence in the retrieved pages is NATIONAL: ITU's Digital Development Dashboard for Egypt reports population covered by at least a 3G mobile network (2022) = 100% and at least a 4G mobile network (2022) = 98%. These are total-population figures with no urban/rural split. The only rural-disaggregated Egyptian f...*

Recorded value: DATA GAP – 1) ITU Statistics landing page – confirms ITU publishes population covered by mobile-broadband network 'total and by urban/rural area', but only as global/regional/development-group aggregates, not per-country. 2) ITU DataHub home and About pages – confirm the World Telecommunication Ind...

- 5. Cost and time. \$6.62** across 629 vendor calls in 15 minutes, against a \$500 country ceiling — 1.3% of it, across 2 passes (EGY\_shadow, EGY\_shadow\_g2). By vendor: openai \$3.92, perplexity \$1.42, exa \$0.74, jina \$0.37, anthropic \$0.17.

## Prerequisites — the twelve rows that gate the matrix

| id  | prerequisite                       | verified   | shadow     | levels              | why the shadow run differs |
|-----|------------------------------------|------------|------------|---------------------|----------------------------|
| 2.1 | Rural mobile broadband coverage (3 | Unverified | Unverified | no level → no level |                            |

| id   | prerequisite                       | verified   | shadow            | levels        | why the shadow run differs                                                                                       |
|------|------------------------------------|------------|-------------------|---------------|------------------------------------------------------------------------------------------------------------------|
| 2.9  | Rural electricity access (%)       | Present    | Present           | L5 → L5       |                                                                                                                  |
| 3.3  | National farmer registry           | Present    | <b>Unverified</b> | L4 → no level | The indicator names a registry of FARMERS (individuals). The only national instrument evidenced in the retrie... |
| 3.11 | Agricultural data interoperability | Absent     | Absent            | L1 → L1       |                                                                                                                  |
| 4.1  | Data protection/privacy law        | Present    | Present           | L3 → L3       |                                                                                                                  |
| 4.5  | Agricultural data governance frame | Absent     | Absent            | L1 → L1       |                                                                                                                  |
| 4.7  | Digital ID coverage (%)            | Present    | Present           | L5 → L5       |                                                                                                                  |
| 4.9  | Inter-ministerial coordination mec | Present    | Present           | L4 → L3       |                                                                                                                  |
| 5.5  | Digital extension capability       | Present    | Present           | L4 → L3       |                                                                                                                  |
| 5.7  | MoAg digital/AI unit               | Unverified | <b>Present</b>    | no level → L3 | A Digital Transformation Unit (AG-DTU) was ordered established inside Egypt's                                    |

| id   | prerequisite                       | verified | shadow            | levels        | why the shadow run differs                                                                                                                             |
|------|------------------------------------|----------|-------------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6.14 | Agri-fintech rails for smallholder | Present  | <b>Unverified</b> | L4 → no level | Ministry of Agriculture and Lan...<br>The indicator names AGRI-FINTECH RAILS for SMALLHOLDERS. The quoted T3 page sits under a heading covering 'NG... |
| 7.12 | Responsible-AI safeguards (consent | Absent   | <b>Present</b>    | L1 → L3       | Egypt adopted a cross-sector Egyptian Charter for Responsible AI, launched by the National Council for Arti...                                         |

## Readiness matrix

| use case | verified   | shadow     |       |
|----------|------------|------------|-------|
| ADV      | Unverified | Unverified | match |
| SMF      | Unverified | Unverified | match |
| MKT      | Unverified | Unverified | match |
| SCM      | Unverified | Unverified | match |
| FIN      | Unverified | Unverified | match |
| AGI      | Blocked    | Blocked    | match |

## Pillars

| pillar | verified mean (band) | rated/held | shadow mean (band) | rated/held |
|--------|----------------------|------------|--------------------|------------|
| A1     | 3.0 (Established)    | 8/2        | 3.6 (Advanced)     | 5/2        |

| pillar | verified mean<br>(band) | rated/held | shadow mean<br>(band) | rated/held |
|--------|-------------------------|------------|-----------------------|------------|
| C1     | 3.8 (Advanced)          | 5/0        | 4.0 (Advanced)        | 4/0        |
| C2     | 2.88<br>(Established)   | 8/0        | 2.8 (Established)     | 5/3        |
| C3     | 3.25<br>(Established)   | 8/0        | 3.29<br>(Established) | 7/1        |
| C4     | 3.33<br>(Established)   | 3/3        | 3.0 (Established)     | 3/1        |
| E1     | 3.11<br>(Established)   | 9/0        | 2.33 (Emerging)       | 6/3        |
| O1     | 2.67<br>(Established)   | 6/0        | 2.0 (Emerging)        | 3/0        |

## Which direction the divergences run

This is the part to read first. A shadow run that withholds a level where the verified assessment set one costs coverage; a shadow run that sets a level *higher* than the verified assessment is claiming readiness the evidence may not carry, and that is the failure that matters.

- **3 rows read higher** than the verified assessment.
  - **2.11 Device financing/subsidy schemes** — L1 to L3. Egypt’s Ministry of Communications and Information Technology operates ProGIG, a joint initiative of the National Telecommunication Institute and Nasser Social Bank offering soft loans of up to EGP 6...
  - **3.8 National soil map/database** — L1 to L3. The indicator names a ‘National soil map/database’. The best available evidence describes (a) national digital soil and terrain data of Egypt produced by NARSS with FAO/European Soil Bureau, and (b) ...
  - **7.12 Responsible-AI safeguards (consent, rights)** — L1 to L3. Egypt adopted a cross-sector Egyptian Charter for Responsible AI, launched by the National Council for Artificial Intelligence in April 2023, built on five values (Human-Centeredness, Transparency an...
- **9 rows read lower.** 2.5 (L5 → L4), 3.6 (L4 → L3), 4.6 (L4 → L3), 4.9 (L4 → L3), 5.5 (L4 → L3), 6.9 (L4 → L1), 3.9 (L4 → L3), 3.10 (L4 → L3), 8.12 (L3 → L1)
- **15 rows withheld a level** the verified assessment set: 1.5, 1.6, 1.7, 2.7, 3.3, 3.5, 3.7, 4.4, 5.8, 6.4, 6.12, 6.14, 8.4, 8.9, 8.11.
- **1 row set a level** the verified assessment withheld: 5.7.

The asymmetry between the last two is the abstention threshold, stated as a number. It is the figure to tune, and tuning it in either direction moves the first bullet — which is the one that decides whether a machine-set readiness matrix can be trusted.

## Where the shadow run withheld a level

Each gate below is a design decision doing its job. A row that reaches a gate keeps its evidence; what it loses is the level, and with it its place in every mean.

| gate             | rows |
|------------------|------|
| construct (hold) | 10   |
| tier (hold)      | 3    |

## Independent corroboration

12 rows are also covered by a machine-fetchable T1 series, fetched separately and never substituted for the research lane's own answer. **7 of 12** research values agree with the independent series within 2%.

## Every row

| id  | indicator                            | verified      | shadow              |                | note                                                                                                   |
|-----|--------------------------------------|---------------|---------------------|----------------|--------------------------------------------------------------------------------------------------------|
| 1.1 | Agriculture value added per worker ( | Measured L4   | Measured L4         | match          |                                                                                                        |
| 1.2 | Cereal yield (kg/ha)                 | Measured L5   | Measured L5         | match          |                                                                                                        |
| 1.3 | Employment in agriculture (%)        | Measured L3   | Measured L3         | match          |                                                                                                        |
| 1.4 | Food production index (2014-16=100)  | Measured L3   | Measured L3         | match          |                                                                                                        |
| 1.5 | Post-harvest loss rate (%)           | Documented L4 | Documented no level | <b>differs</b> | The indicator name asserts a single all-crop NATIONAL post-harvest loss rate for Egypt. No such fig... |
| 1.6 | Smallholder access to formal markets | Documented L1 | Gap no level        | <b>differs</b> | Worked through the ten supplied pages in order. Sources 1, 2,                                          |

| id                                                                                      | indicator                            | verified            | shadow              |                | note                                                                                                                                               |
|-----------------------------------------------------------------------------------------|--------------------------------------|---------------------|---------------------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.7                                                                                     | Agricultural credit access (% farmer | Documented L1       | Gap no level        | <b>differs</b> | 3 and 4 (IFAD, T1): the 2017 CSPE, th...<br>1) FAO Statistics highlight 'Credit to agriculture. Global and regional trends 2014-2023' — checked... |
| 1.8                                                                                     | Farmers using climate-smart practice | Documented no level | Gap no level        | match          |                                                                                                                                                    |
| 8.1                                                                                     | Prevalence of undernourishment (%)   | Measured L3         | Measured L3         | match          |                                                                                                                                                    |
| 8.5                                                                                     | Women who own land (% holders)       | Measured no level   | Documented no level | match          |                                                                                                                                                    |
| 2.1  | Rural mobile broadband coverage (3G/ | Gap no level        | Gap no level        | match          |                                                                                                                                                    |
| 2.4                                                                                     | Individuals using the Internet (%)   | Measured L4         | Measured L4         | match          |                                                                                                                                                    |
| 2.5                                                                                     | Mobile broadband price (% GNI pc)    | Measured L5         | Measured L4         | <b>differs</b> | 1.58                                                                                                                                               |
| 2.7                                                                                     | Rural smartphone ownership (%)       | Measured L4         | Gap no level        | <b>differs</b> | 1) ITU DataHub Egypt page for household indicators with Location disaggregation — only                                                             |

| id                                                                                     | indicator                            | verified      | shadow              |                | note                                                                                                   |
|----------------------------------------------------------------------------------------|--------------------------------------|---------------|---------------------|----------------|--------------------------------------------------------------------------------------------------------|
|                                                                                        |                                      |               |                     |                | 'Households ...                                                                                        |
| 2.9   | Rural electricity access (%)         | Measured L5   | Measured L5         | match          |                                                                                                        |
| 2.11                                                                                   | Device financing/subsidy schemes     | Documented L1 | Documented L3       | <b>differs</b> | Egypt's Ministry of Communications and Information Technology operates ProGIG, a joint initiative o... |
| 3.1                                                                                    | UN E-Government Development Index    | Measured L4   | Measured L4         | match          |                                                                                                        |
| 3.3  | National farmer registry             | Documented L4 | Documented no level | <b>differs</b> | The indicator names a registry of FARMERS (individuals). The only national instrument evidenced in ... |
| 3.4                                                                                    | Digital land/plot registration       | Documented L3 | Documented L3       | match          |                                                                                                        |
| 3.5                                                                                    | Open agricultural data (machine-read | Documented L2 | Documented no level | <b>differs</b> | The indicator names OPEN + MACHINE-READABLE + AI-READY AGRICULTURAL data. The strongest artifact fo... |

| id     | indicator                                  | verified         | shadow                 |                | note                                                                                                   |
|--------|--------------------------------------------|------------------|------------------------|----------------|--------------------------------------------------------------------------------------------------------|
| 3.6    | Weather/<br>climate data<br>infrastructure | Documented<br>L4 | Documented<br>L3       | <b>differs</b> | The Egyptian Meteorological Authority operates 28 synoptic stations (all shared on the WMO Global T... |
| 3.7    | Satellite/EO<br>data<br>integration        | Documented<br>L4 | Documented<br>no level | <b>differs</b> | The indicator name, 'Satellite/EO data integration', fixes neither the object (integration into wha... |
| 3.8    | National soil<br>map/database              | Documented<br>L1 | Documented<br>L3       | <b>differs</b> | The indicator names a 'National soil map/database'. The best available evidence describes (a) natio... |
| 3.11 ► | Agricultural<br>data<br>interoperability s | Documented<br>L1 | Documented<br>L1       | match          |                                                                                                        |
| 4.1 ►  | Data<br>protection/privacy law             | Documented<br>L3 | Documented<br>L3       | match          |                                                                                                        |
| 4.2    | Cybersecurity framework<br>(ITU GCI)       | Measured L5      | Measured L5            | match          |                                                                                                        |
| 4.3    | Government<br>AI Readiness<br>Index        | Measured L3      | Measured L3            | match          |                                                                                                        |
| 4.4    | National<br>digital<br>agriculture         | Documented<br>L1 | Documented<br>no level | <b>differs</b> | The indicator names a 'National                                                                        |

| id                                                                                      | indicator                            | verified            | shadow        |                | note                                                                                                   |
|-----------------------------------------------------------------------------------------|--------------------------------------|---------------------|---------------|----------------|--------------------------------------------------------------------------------------------------------|
|                                                                                         | strateg                              |                     |               |                | digital agriculture strategy'. The best evidence retrieved is for a...                                 |
| 4.5    | Agricultural data governance framewo | Documented L1       | Documented L1 | match          |                                                                                                        |
| 4.6                                                                                     | National AI strategy with agricultur | Documented L4       | Documented L3 | <b>differs</b> | Egypt's National AI Strategy contains a dedicated agriculture component: Section 7.1 'Agriculture, ... |
| 4.7   | Digital ID coverage (%)              | Measured L5         | Measured L5   | match          |                                                                                                        |
| 4.9  | Inter-ministerial coordination mecha | Documented L4       | Documented L3 | <b>differs</b> | Under the guidance of the Inter-Ministerial Steering Committee, UELDP enhanced Local economic devel... |
| 5.2                                                                                     | Adult literacy rate (%)              | Measured L3         | Measured L3   | match          |                                                                                                        |
| 5.3                                                                                     | Tertiary STEM enrollment (% gross)   | Measured no level   | Gap no level  | match          |                                                                                                        |
| 5.4                                                                                     | Digital literacy among farmers (%)   | Documented no level | Gap no level  | match          |                                                                                                        |

| id                                                                                    | indicator                            | verified            | shadow          |                | note                                                                                                   |
|---------------------------------------------------------------------------------------|--------------------------------------|---------------------|-----------------|----------------|--------------------------------------------------------------------------------------------------------|
| 5.5  | Digital extension capability         | Documented L4       | Documented L3   | <b>differs</b> | In July 2019 FAO and Egypt's Ministry of Agriculture and Land Reclamation launched a digital model ... |
| 5.7  | MoAg digital/AI unit                 | Documented no level | Documented L3   | <b>differs</b> | A Digital Transformation Unit (AG-DTU) was ordered established inside Egypt's Ministry of Agricultu... |
| 5.8                                                                                   | Agtech/data-science training pipelin | Documented L3       | Judged no level | <b>differs</b> | the only source is T5 news, vendor or market material, which carries existence facts but never a st... |
| 5.12                                                                                  | Gender balance in digital-ag workfor | Gap no level        | Gap no level    | match          |                                                                                                        |
| 6.1                                                                                   | Global Innovation Index              | Measured L1         | Documented L1   | match          |                                                                                                        |
| 6.3                                                                                   | Business Ready (B-READY)             | Gap no level        | Gap no level    | match          |                                                                                                        |
| 6.4                                                                                   | Agtech venture ecosystem             | Documented L3       | Judged no level | <b>differs</b> | the only source is T5 news, vendor or market material,                                                 |

| id     | indicator                            | verified      | shadow              |                | note                                                                                                                                                      |
|--------|--------------------------------------|---------------|---------------------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6.9    | Public-private partnerships in digit | Documented L4 | Documented L1       | <b>differs</b> | which carries existence facts but never a st...<br>Across the pages retrieved for Egypt I found digital-agriculture collaborations that are public-pub... |
| 6.12   | Digital public goods adopted         | Documented L4 | Judged no level     | <b>differs</b> | the only source is T5 news, vendor or market material, which carries existence facts but never a st...                                                    |
| 6.13   | SME/agribusiness adoption of digital | Gap no level  | Gap no level        | match          |                                                                                                                                                           |
| 6.14 ► | Agri-fintech rails for smallholders  | Documented L4 | Documented no level | <b>differs</b> | The indicator names AGRI-FINTECH RAILS for SMALLHOLDERS. The quoted T3 page sits under a heading co...                                                    |
| 3.9    | Digital advisory platforms at scale  | Documented L4 | Documented L3       | <b>differs</b> | Digital agricultural advisory apps, websites, call centers and SMS platforms                                                                              |

| id                                                                                     | indicator                            | verified      | shadow        |                | note                                                                                                                             |
|----------------------------------------------------------------------------------------|--------------------------------------|---------------|---------------|----------------|----------------------------------------------------------------------------------------------------------------------------------|
| 3.10                                                                                   | Agricultural e-commerce platforms    | Documented L4 | Documented L3 | <b>differs</b> | (e.g. Hodhod, Mahsooly... Farmer-facing agricultural e-commerce/online marketplace platforms are operating in Egypt, includin... |
| 7.2                                                                                    | AI-enabled agricultural solutions de | Documented L3 | Documented L3 | match          |                                                                                                                                  |
| 7.12  | Responsible-AI safeguards (consent,  | Documented L1 | Documented L3 | <b>differs</b> | Egypt adopted a cross-sector Egyptian Charter for Responsible AI, launched by the National Council ...                           |
| 8.2                                                                                    | Account ownership, female (%)        | Measured L3   | Measured L3   | match          |                                                                                                                                  |
| 8.4                                                                                    | Mobile money account (%)             | Measured L2   | Gap no level  | <b>differs</b> | 1) Checked Source 1 (data.worldbank.org, Egypt) first as the canonical T1 route — found only FX.OWN...                           |
| 8.6                                                                                    | Gender gap in phone ownership (pp)   | Measured L2   | Documented L2 | match          |                                                                                                                                  |

| id   | indicator                            | verified      | shadow        |                | note                                                                                                   |
|------|--------------------------------------|---------------|---------------|----------------|--------------------------------------------------------------------------------------------------------|
| 8.9  | Smallholders reached by digital serv | Documented L1 | Gap no level  | <b>differs</b> | Reviewed all ten supplied pages in order of likely relevance to a national smallholder reach figure... |
| 8.11 | Services in local languages (%)      | Documented L5 | Gap no level  | <b>differs</b> | 1) Checked MCIT press release ‘ICT Sector Achievements in 2025’ (T3) — the most granular Egyptian g... |
| 8.12 | Documented impact evidence (yield/in | Documented L3 | Documented L1 | <b>differs</b> | Across the retrieved Egypt evidence base — IFAD’s three country strategy and programme evaluations ... |
| 8.17 | Climate advisory reach (%)           | Gap no level  | Gap no level  | match          |                                                                                                        |

■ marks a prerequisite.

## Reading this

Divergence here is the expected result. The verified assessments came from sustained human-directed searching under the full tiered protocol — Nigeria went from 21 recorded gaps to 4 that way — and this pass runs once, on a budget, without the Gate 2 refutation round that found four of those gap refutations. More gaps and more holds are the honest output of a first automated pass, not a regression.

The number to act on is the **abstention rate**: 10 holds and 14 gaps against the verified 5 and 5. Too loose and everything reads Ready; too tight and everything reads Unverified. These figures

are what that threshold should be tuned against, and they should be kept — when automated Gate 2 arrives, re-running this comparison is what tells you whether it earns its 15% of the budget.